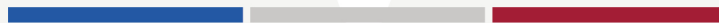




MIT SLOAN SCHOOL OF MANAGEMENT
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MODULE 1 UNIT 4
Podcast 1 Transcript

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The room and the question

PAUL MCDONAGH-SMITH: On a Friday morning in early March, on the second floor of a brick building in the center of Birmingham, six people sat around a table and tried to answer a question that most of them were not yet sure how to ask. The building belongs to Boulton Industries, a diversified industrial group with roots going back more than a century, operations across manufacturing, logistics, and professional services, and about 4,000 employees spread across three continents. The table is in the executive meeting room. The people around it are the senior leadership of the company, the Chief Executive, the Chief Operating Officer, the Chief Financial Officer, the Chief Technology Officer, the General Counsel, and the Chief People Officer. Six decision-makers, one question and three weeks on the clock.

I'm going to tell you what I heard in that room and in the two rooms that followed it on the two Fridays thereafter. I was there as an observer. I had been asked to sit in, to take notes, and at the end of the three weeks, help the team turn their reasoning into a document that the board would be able to read.

And so, my name is Paul McDonagh-Smith. I teach executive education at MIT Sloan School of Management. What follows is my account of these three meetings, during which a capable industrial company tried to work out what agentic AI actually is, what they should do about it, and most difficult of all, what kind of organization they would need to become in order to deploy it responsibly.

The question the board had put to them four weeks earlier was simple to state and hard to answer: "Given the pace and scale of developments in agentic AI, produce a recommendation to the board within three weeks on how Boulton Industries should respond." No more, no less. The board wasn't asking whether agentic AI was real. It was asking what Boulton Industries would do about it.

The Chief Executive, who had been in the role for six years and had led the company through its previous wave of digital transformation, opened the meeting by reading the board's remits aloud. And then she put the paper down and said something that surprised me. "Before we decide what Boulton should do," she said, "I'd like each of us to say in one sentence what we mean when we use the word 'agent.' I'll start, I mean a piece of software that takes an instruction and turns it into a sequence of actions in the world without a person supervising every step." She looked around the table. "One sentence each, your turn."

Now, I want to pause the tape here, so to speak. Because what happened in the next three minutes, set the shape of everything that followed.

Five definitions of one word

PAUL MCDONAGH-SMITH: The Chief Technology Officer, Vikram Sen, six months into a newly created role at the company, previously a research engineer in the automotive sector, went next. "An agent," he said, "is a large language model wrapped in a system

that lets it call external tools, keep track of what it's done, and reason about what to do next."

The General Counsel, Olu Adedayu, who had joined Boulton three years earlier from financial services regulation, frowned slightly. "That's a technical definition. Mine is different. An agent is a software actor with some degree of decision-making authority delegated to it. The question I care about is how much authority, under what conditions, and with what accountability."

And then the Chief Financial Officer, Rebecca Holloway, the longest serving member of the leadership team, went next. "Mine is narrower," she said. "An agent is a system I can assign work to and expect a result from. Where the work involves multiple steps and some judgement, and where I get back either a completed deliverable or a clear escalation."

The Chief Operating Officer, Maria Santos, who ran the three manufacturing facilities and the logistics network, gave what I considered a particularly practical answer. "An agent is whatever does the work that would otherwise have gone into a team's queue. If it clears the queue, it's an agent. If it doesn't, it's a fancy autocomplete."

And the Chief People Officer, Priya Kapoor, who carried workforce, culture, and organizational development, gave the answer that, in retrospect, did the most work. "An agent," she said, "is something my people are going to have to learn to manage; not use, to manage. That's the definition I'm holding in my head."

So, six people, five substantively different definitions. And all of them, as far as I could tell, were defensible. The CEO had let each person finish without interrupting. Then, when Priya sat back, she looked around the table and said quite calmly, "That's our first problem." Here is what I think she'd seen and what I want to surface for you. The word "agent" is what the MIT professor Marvin Minsky would have termed a "suitcase word" – that is, a term we pack with a number of different meanings and carry around with us as if it were one thing. Minsky was writing about words like "intelligence" and "consciousness". But the observation generalizes. Words we share the use of, without sharing the meaning of, can become organizational hazards. They feel like alignment, but they aren't.

Around the Boulton Industries table, each definition carried a different decision-making frame behind it. Vikram's technical definition pointed towards an architecture discussion. Olu's authority and accountability definition pointed towards governance. And Rebecca's work unit definition pointed toward commercial value. And Maria's queue clearing definition pointed towards operational deployments. And then Priya's something my people must learn to manage definition pointed clearly toward the organizational change that would be required to absorb the technology. Five definitions, five strategies in waiting. Each person around the table thought they were agreeing with the others. And they were, at the level of words, but they weren't at the level of meaning.

The CEO wasn't flustered by this. Indeed, I got the impression she had somewhat expected it. What she said next was, in effect, the theses of this first meeting: "Before we can tell the board what to do, we have to tell each other what we're talking about. And the fact that we just used the same word five different ways tells me we haven't yet earned the right to make a recommendation. So, let's spend the rest of today making sure we understand the terrain honestly and clearly together. We'll spend the next week on what to build. We'll

spend the week after on what to do about it at scale.” And she said, “Today is about vocabulary.”

And that I wrote in my notebook. It’s how the first meeting properly began.

The continuum, and what changed

PAUL MCDONAGH-SMITH: Vikram, the Chief Technology Officer, had brought a single slide. He put it up on the screen at the end of the room. It was a horizontal timeline from left to right with five points: classical machine learning, deep learning, generative AI, agentic AI, and artificial general intelligence. Each point had a brief descriptor beneath it. The gap between generative AI and agentic AI was visibly wider than the others. Vikram walked the room through it.

Classical machine learning, he said, was about learning patterns in structured data. The sort of work Bolton was already doing in predictive maintenance and demand forecasting. Deep learning had extended that capability to unstructured inputs, to images, to audio, to natural language. And then generative AI had crossed a further threshold, producing novel outputs rather than merely classifying existing ones. “But every version of AI we’ve deployed at Bolton,” he said, “including generative AI, is reactive. You give it an input. It gives you an output. You give it a question, and it gives you an answer. However sophisticated, the pattern is the same. Prompt in, response out.”

He paused. “Agentic AI is different. It isn’t a faster version of what we’ve already done. It’s a qualitative shift, not a quantitative one. You give it a goal, it figures out the steps, it chooses which tools to use. It executes. It evaluates. It adjusts. And this is the part that matters. It takes action without you supervising each step.”

And then Rebecca, the CFO, asked him to say that last part again. And he did. “Prompt in, response out,” Vikram said, “Versus goal in, plan, execute, adapt out. He then gave what I’ve since heard described as a useful analogy for the shift. And I want to give it to you here in the form he used: “If a generative AI is like an expert witness in a courtroom,” Vikram said, “then an agentic AI is the project manager running the case. The expert witness is brilliant when asked the right question. But they don’t run the case. They don’t decide which witnesses to call, in what order, or when to change strategy. A project manager does. That’s the shift from responding to pursuing.”

The analogy did work in the room. I watched it land. Rebecca took notes. Olu nodded. Maria said, “Ah, that’s helpful. That’s the first framing I’ve heard that explains what’s actually different.” And then Vikram said something that a researcher he’d met at MIT had told him, and which he had started using as a piece of shorthand: “Agentic AI uses the same technology family as what we already have, but it’s a different species.” There was a silence. The chief executive, who had been listening rather than speaking, said quietly, “Good, good. Then we need to treat it as a different species. Which means our existing AI strategy doesn’t automatically cover it. Which means we, at this table, are behind.”

Now, I should mention something here that the Boulton Industries teams would return to throughout the next three weeks. Vikram’s slide had a sixth column, off to the right of Artificial General Intelligence, that was deliberately left blank. When Rebecca asked him what it was for, he said, “It’s for what we haven’t yet named. The continuum doesn’t stop moving. Whatever we decide today will need to be revisable.” That remark, more than

anything else said that morning, captured the temperament the team were trying to bring to their work. Not “Let us decide this and be done,” but “Let us decide this in a way that can be revised.”

Three waves, and where Boulton sits

PAUL MCDONAGH-SMITH: After a short break, Priya, the Chief People Officer, said she wanted to bring in a framing she’d been carrying around for a couple of weeks. She had been reading what she described as “the practitioner literature,” the material produced by people deploying enterprise AI at scale. And she had come across a three-wave model that she thought made the Boulton Industries’ situation legible.

The first wave, she said, had been about virtual agents, chatbots, customer service deflection, simple automations. Most large organizations, Boulton Industries included, had been through this wave some years ago. The ambition was modest. The technology was accessible. The wave had crested and left a residue of useful capability in its wake. The second wave had been about internal productivity. Organizations bringing their own data together, letting their employees query it, draft with it, summarize with it. Boulton was, by any fair assessment, still in the middle of this wave. Some of its functions were fluent; others had barely begun.

“And the third wave,” Priya said, “is the one we’re being asked about today. It’s about systems that do more than generate content for people to review. It’s about systems that act, that complete work, that execute. This is the wave that the board wants us to have a view upon.” And then she said something that I underlined in my notebook: “The temptation in a moment like this is to treat wave three as the only one that matters now. To redirect resources, chase the agentic frontiers, declare ourselves an agent-first company, and let the second wave take care of itself. I want to put on the record now that I think that would be a mistake.”

Rebecca asked her why. “Because,” Priya said, “wave three only pays off if wave two is mature. Agentic systems take action if the organization underneath them isn’t already thinking and acting with AI as a default. For example, if the data doesn’t flow, if the processes haven’t been redesigned, if the people haven’t been developed, then you end up grafting autonomous systems onto an organization that can’t readily absorb them. And when that happens, the agents don’t lift the organization, they potentially break it.”

She then offered a distinction I thought was a useful piece of language introduced in this first meeting. “There is a difference,” she said, “between being AI-first and being AI-native. An AI-first organization prioritizes AI in its strategy documents, its investment decisions, its external communications. An AI-native organization thinks and acts with AI as its default mode of work. The first is a statement. The second is a culture. Most enterprises are AI first. Very few are AI native.”

“And the agentic wave,” Priya said, “requires you to be more AI native before it can really begin to deliver value.” Now, Maria, who had been quiet during most of the continuum discussion, came back into the conversation at this point. “Alright,” she said, “let’s locate ourselves honestly. Where are we on each of these three waves?” The CEO agreed that the exercise was worth doing. And, for the next forty minutes or so, the team worked through it together. I won’t reproduce the detail because some of it’s specific to Boulton Industries. But the shape of the conclusion was this: They rated themselves as strong on

wave one, competent but incomplete on wave two, and ambitious but unprepared on wave three.

Ambitious but unprepared. That phrase captured the problem precisely. And I watched it travel around the room. Rebecca wrote it on the whiteboard. Olu nodded slowly. Vikram, who I think had walked in that morning expecting the meeting to end with a mandate to accelerate, visibly adjusted. The CEO said, “So, the recommendation to the board can’t simply be let’s do agentic AI. It has to be let’s do agentic AI in a way that simultaneously accelerates our wave two maturity. Because without that wave three simply won’t work.”

Now, I consider this to be probably the most important sentence spoken in that first meeting. It reframed the entire recommendation from a technology decision into an organizational one.

The three realities, and the question they left on the table

PAUL MCDONAGH-SMITH: The last hour of this first meeting was spent on something the CEO called in her opening remarks, “the realities we ought to put on the table before we go any further.” She had written three of them down on an index card before the meeting started. I saw her glance at it once and she offered them as a way of closing the morning honestly.

The first reality the speed of iteration matters more than the perfection of the plan. The landscape was moving quickly enough that a long evaluation cycle was in itself a form of risk. A six-month study to pick a platform could, Vikram noted, produce a recommendation that was already obsolete by the time it was signed. The implication for Boulton Industries wasn’t that they should move recklessly, but that they needed to build a capacity for rapid prototyping and tight feedback loops, even into the board recommendation itself.

And then the second reality: The biggest barriers are organizational, not technical. This was the reality Priya had been circling all morning. The evidence from the wider industry, she said, was consistent. Projects that failed did so overwhelmingly for organizational reasons: misaligned incentives, unclear accountability, data that wouldn’t flow between systems that had never been designed to talk to one another. The models were capable; the organizations weren’t. Rebecca agreed and added that, in her experience, the most expensive technology was almost always the technology you bought before you had the process to use it.

The third reality was that existing metrics alone won’t capture the value agentic systems create. When an agent resolves a customer issue end to end, Rebecca said, the value isn’t only in the minutes saved, it’s in the strength or the weakness of the lifetime relationships that follow. When an agent completes a piece of work that a human would otherwise have done, the value is not in the labor cost reclaimed. It’s in the capacity to pursue work you would previously have declined. Boulton Industries’ existing measurement frameworks, she said plainly, were not yet fit for this.

And so, these three realities became, for the remainder of the next three weeks, a kind of shared reference point. The team returned to them repeatedly. Now, I want to name them here, because they are the frame inside which the rest of the Boulton conversation took place.

The meeting closed a little before noon. The CEO summarized. “We came in with the board’s question,” she said. “Let me try to restate it now, having spent the morning together. The question is not ‘Should Boulton Industries adopt agentic AI?’ That question is behind us. The question is ‘What kind of organization does Boulton Industries need to become in order to deploy agentic AI well?’ And what is the first decision this year that moves us meaningfully in that direction?” And she said, “Next Friday, we’ll bring a concrete proposal for a first agentic deployment. We’ll work through what it would take to build it responsibly, what we would need to have in place before we began, and what we would need to learn before we could scale it.”

“The Friday after that,” she said, “we’ll work through what scaling would actually require. And by the end of the three weeks, we’ll have a recommendation the board can receive.” So, she summarized with, “Between now and next Friday, I’d like each of us to bring one thing, something specific, a workflow, a process, a function that we think is a good candidate for Boulton Industries’ first serious agentic AI deployment. Don’t bring the safest one. Don’t bring the most ambitious one. Bring the one that will teach us the most.” And with that, she closed the first meeting. The team dispersed.

I stayed behind for a moment looking at the whiteboard: ambitious but unprepared. The phrase was still there in Rebecca’s handwriting, underlined twice.

Key takeaways

PAUL MCDONAGH-SMITH: Now, let me offer some observations as the person who sat through that morning, and has now had the time to reflect upon it. What I witnessed in that room was a group of capable executives doing something rarer than I had expected. They resisted the temptation to move too quickly to a decision. They interrogated their own language before they interrogated the technology. They acknowledged that their organization was not yet where it would need to be. And they reframed the question they had been given into the question they thought was the right one.

None of that is or was inevitable. I’ve watched other rooms at other companies rush past each of those steps. If I were to compress what the Boulton team learned during the vocabulary week into a short set of propositions, it would be these:

- Agentic AI is not a faster version of the AI you already have. It’s a qualitative shift from systems that respond to systems that pursue.
- Your organization is probably riding more than one wave of AI adoption at once. And pretending or considering that you can skip the earlier waves, when in fact that will cost you when the later ones arrive.
- The binding constraint on Agentic AI deployment is almost never technical. It’s organizational.
- The difference between being AI first and AI native is the difference between a strategy and a culture. And agentic AI requires the second.
- And then the metrics you already use will most likely fail capture the value or all of the value or the risk that agentic systems create.

These propositions didn't come from a textbook. They came from five senior executives sitting in a room in Birmingham, trying in good faith to tell each other the truth about where their organization stood. The meeting had set up the question. The next meeting, the following Friday, would bring the first proposal. I'll tell you what happened in that room next time.